

16 Thermodynamics

A · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Internal energy is the total random microscopic kinetic and potential energy of particles. First law: $\Delta U = Q + W$ when W is work done on the system (check sign convention). During melting at constant temperature, energy increases molecular potential energy. Specific heat capacity relates heating to $mc\Delta T$; latent heat relates phase change to mL .

Worked example: follow the reasoning

Heating 0.50 kg of water by 10 K with $c = 4200 \text{ J kg}^{-1} \text{ K}^{-1}$ requires $Q = 0.50 \times 4200 \times 10 = 21 \text{ kJ}$ if losses are neglected.

Quick check · multiple choice

A A gas receives heat Q and does work W_{out} . What is ΔU ?

[1]

A $Q + W_{\text{out}}$

B $W_{\text{out}} - Q$

C $-Q - W_{\text{out}}$

D $Q - W_{\text{out}}$

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 A gas receives 400 J of heating and does 150 J of work on its surroundings. Find ΔU and explain the sign of the work term.

[3]

Final answer: _____

Answers and explanation

Quick check: D. $Q - W_{\text{out}}$

Question 1 [3 marks]

Work done on gas = -150 J [1]. $\Delta U = Q + W = 400 - 150 = +250 \text{ J}$ [1]; expansion transfers energy out of the gas [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Physics 9702, 2025–2027. <https://www.cambridgeinternational.org/Images/664565-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.